

FITFORGE

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Summit 350 S Treadmill

SERVICE MANUAL

FF-TT-SUMMIT-350-S



Illustration shows the machine in the running position. Specifications may vary by market.

This manual is intended for use by the equipment owner and by FitForge authorised service technicians.
Read Section 2 in full before carrying out any procedure.

Document FF-SM-IT-350-S · Revision C · Model year 2021 · Treadmill

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SECTION 1 — MODEL IDENTIFICATION

Confirm the model before ordering parts or following any procedure in this manual. Procedures and part numbers differ between models in the same range, and fitting the wrong part will not resolve the fault.

SERIAL PLATE LOCATION
Underside of the frame, on the front stabiliser bar.
Fold the deck up and look down the left-hand rail.

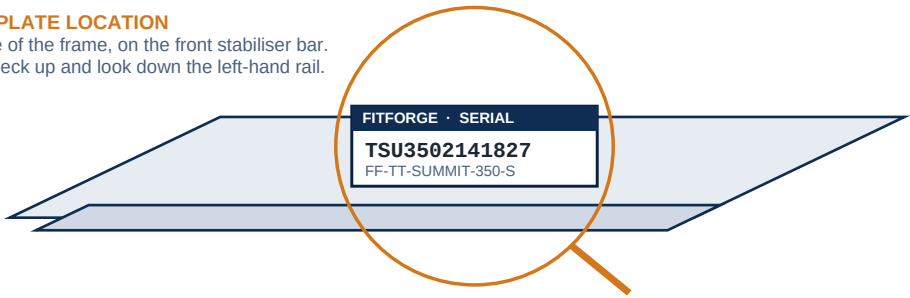


Figure 1 — Serial plate location

The plate is fixed on the frame beneath the motor hood, at the front of the machine.
Serial numbers for this model begin with **TSU350**, followed by a two-digit year and a five-digit sequence.

If the plate is missing or unreadable, the following characteristics distinguish this model from others in the range:

Characteristic	This model
Deck	charcoal
Console	7in colour touch
Folding	folding
Incline	-3 to 15%

SECTION 2 — SAFETY

Read this section completely before operating or servicing the equipment. The procedures in this manual assume the machine is unplugged unless a step explicitly states otherwise.

⚠ DANGER — MAINS VOLTAGE

This equipment contains mains-voltage components. Never remove the motor hood, the controller cover, or any panel secured with tamper-resistant screws.

Always unplug the unit at the wall before any inspection or maintenance procedure described in this manual.

If you smell burning, see smoke or sparks, or the circuit breaker trips repeatedly, unplug the unit immediately and contact FitForge service. Do not attempt to diagnose the fault yourself.

⚠ CAUTION — MOVING PARTS

Keep children and pets away from the equipment at all times, including when it is not in use. Remove the safety key and store it out of reach.

Keep hands, feet, loose clothing and hair clear of the belt and rollers while the unit is powered.

Position the equipment on a level surface with at least 2 m of clear space behind it and 0.6 m to each side.

General requirements

- Always attach the safety key clip before starting a workout. The belt will not start without it.
- Keep children and pets at least two metres from the machine while it is in use.
- Inspect the equipment before every use. Do not use it if any component is damaged, worn or missing.
- Do not exceed the maximum user weight given in Section 3.
- Connect only to a dedicated, earthed 220-240 V outlet. Do not use an extension lead or a multi-way adaptor.
- Allow the motor to cool for 10 minutes before servicing after use.

Safety decal locations

Replace any decal that is missing, torn or illegible before returning the machine to service. Order replacements by position reference.

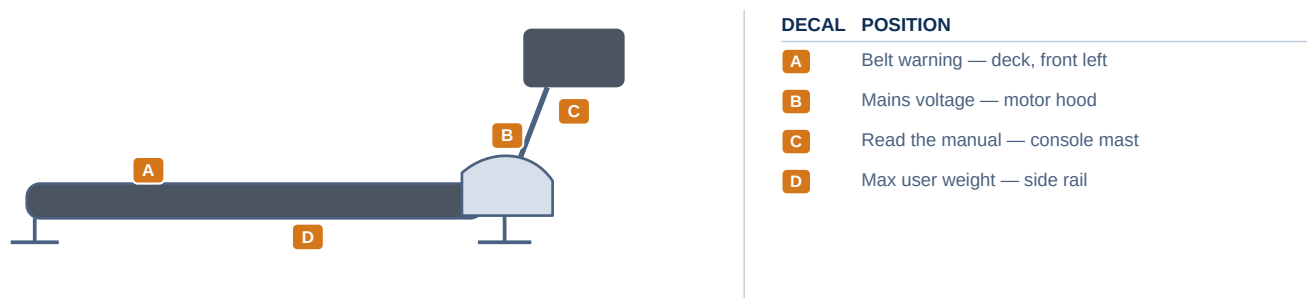


Figure 2 — Safety decal locations

SECTION 3 — SPECIFICATIONS

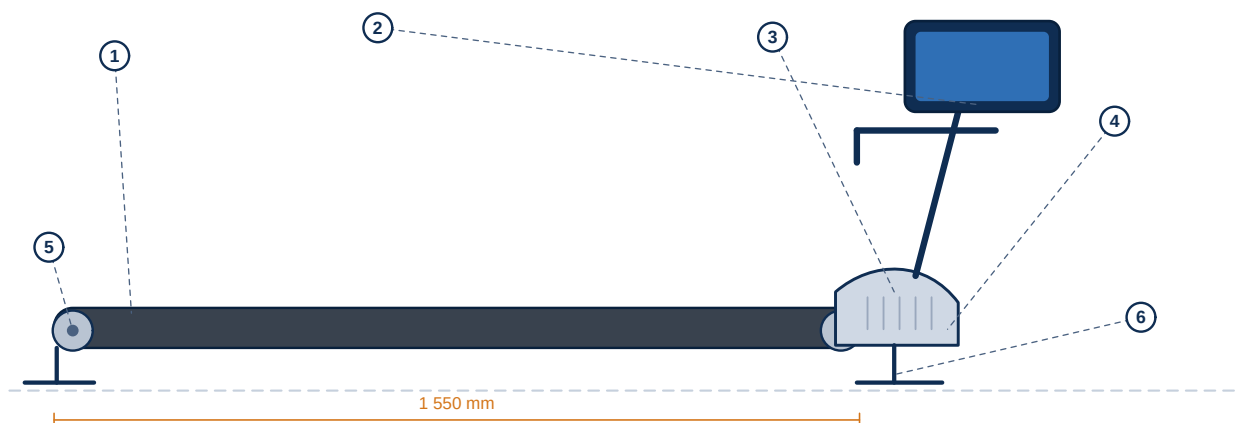


Figure 3 — Principal assemblies

#	Assembly	Service note
1	Running belt and deck	Wear item. Inspect monthly; lubricate every 3 months.
2	Console assembly	Customer replaceable. Disconnect power before removal.
3	Motor hood	Do not remove. Mains voltage inside. Technician only.
4	Drive motor	Technician only.
5	Rear roller and tension bolts	Adjust for belt tension and tracking.
6	Incline actuator	Calibrate from the service menu.

Technical data

Parameter	Value
Running surface	500 mm x 1 550 mm
Motor	3.0 CHP continuous duty
Speed range	0.8 - 20.0 km/h
Incline range	-3 to 15%
Console	7in colour touch
Deck finish	charcoal
Folding	Yes
Maximum user weight	160 kg
Unit weight	118 kg
Supply	220-240 V ~ 50 Hz, dedicated earthed circuit

Torque specifications

Fastener	Size	Torque
Rear roller tension bolts	M8	Quarter-turn increments only
Front roller mounting	M8	22 Nm
Motor mounting bolts	M10	40 Nm
Upright to frame	M10	45 Nm

Fastener	Size	Torque
Console to upright	M6	9 Nm
Deck mounting	M8	24 Nm

 CAUTION - BELT TENSION

Rear roller tension bolts are adjusted in quarter-turn increments and are not torque-specified. Never exceed one full turn in total from the factory setting; over-tensioning destroys the roller bearings and voids coverage on the roller.

SECTION 4 — MAINTENANCE SCHEDULE

The majority of service calls are caused by missed maintenance. Following this schedule prevents most of the faults described in Section 5.

Interval	Task
Every use	Wipe the belt and deck. Check the safety key is present and undamaged.
Weekly	Vacuum around and beneath the unit. Confirm the belt is running centred.
Monthly	Check belt tension. Inspect the power cord along its full length.
Every 3 months / 250 km	Lubricate the deck with FitForge silicone lubricant. Do not use oil.
Every 6 months	Check all frame fasteners against the torque table in Section 3.
Every 12 months	Inspect the drive belt for glazing or cracking. Inspect roller bearings.

Lubrication procedure

1. Unplug the unit at the wall and remove the safety key.
2. Lift one edge of the belt at the centre of the deck.
3. Apply 15 ml of silicone lubricant along the centre line of the deck, reaching about 300 mm forward and back from the middle.
4. Repeat on the opposite side.
5. Refit the safety key, run the belt at 4 km/h for three minutes without anyone on it, then wipe away any excess.

SECTION 5 — TROUBLESHOOTING

Work through the checks for a symptom in the order given. They are ordered cheapest and safest first, and a later check often assumes an earlier one has been completed. Do not skip ahead.

Symptom: belt slipping or hesitating under foot

Also described as: belt slips, belt hesitates, belt catches, jerky belt.

1. Confirm the treadmill is on a level, hard surface and not on carpet or a mat that bunches.
2. With the unit OFF and unplugged, lift the belt at the centre of the deck. You should be able to raise it 50-75 mm (2-3 in). More than that means the belt is loose.
3. If the belt is loose, tighten both rear roller bolts by a quarter turn each, alternating sides, then re-check. Never exceed one full turn total.
4. Walk on the belt at 3 km/h and confirm the slipping has stopped.
5. If the belt is correctly tensioned but still slips, check the drive belt under the motor hood for glazing or cracking.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: belt.

Symptom: belt drifting to one side

Also described as: belt off centre, belt tracking, belt moves left, belt moves right.

1. Run the belt at 5 km/h with nobody on it and note which side it drifts toward.
2. If it drifts RIGHT, turn the right rear roller bolt clockwise a quarter turn.
3. If it drifts LEFT, turn the left rear roller bolt clockwise a quarter turn.
4. Let the belt run for two minutes after each adjustment before judging the result.
5. If more than two full turns are needed to centre the belt, the rear roller is bent or the frame is out of square.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: rear roller.

Symptom: console is blank or will not power on

Also described as: no display, dead console, screen black, unit will not turn on.

WARNING

If the breaker trips repeatedly, or you smell burning, unplug the unit and stop. Do not open the motor hood.

1. Confirm the unit is plugged directly into a wall outlet, not an extension lead or surge protector.
2. Check the reset breaker beside the power inlet — press it fully in if it has tripped.
3. Confirm the safety key is seated in its holder.
4. Try a different wall outlet that you know works.
5. If the breaker trips again immediately after resetting, stop and do not retry — this indicates a short circuit inside the unit.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: power cord, safety key, console.

Symptom: loud squeaking or grinding noise while running

Also described as: *squeak, grinding, screeching, rubbing noise.*

1. Determine whether the noise follows belt speed (roller or belt) or is constant (motor or fan).
2. With the unit off and unplugged, check the deck surface under the belt for a dry, dusty feel — a properly lubricated deck feels slightly slick.
3. If the deck is dry, apply silicone lubricant per the maintenance section.
4. If the noise is a rhythmic squeak once per belt revolution, inspect the belt seam for lifting.
5. If the noise is metallic and constant, the roller bearings have failed.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: belt lube, front roller, rear roller.

Symptom: burning smell or smoke from the unit

Also described as: *burning, smoke, smells hot, electrical smell.*



WARNING

Mains-voltage fault. This must be handled by an authorised technician; no customer-guided diagnosis is permitted.

1. STOP. Switch the unit off at the wall and unplug it immediately.
2. Do not restart the unit to reproduce the symptom.
3. Do not remove the motor hood.

If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: motor, motor controller.

Symptom: incline will not move or is stuck

Also described as: *incline stuck, incline not working, wont incline.*

1. Run the incline calibration routine: hold the STOP and SPEED-UP keys for five seconds with the safety key inserted.
2. Watch whether the incline motor attempts to move at all — listen for a hum.
3. If there is a hum but no movement, the incline actuator is jammed or its nut has stripped.
4. If there is no hum, the controller is not driving the actuator.

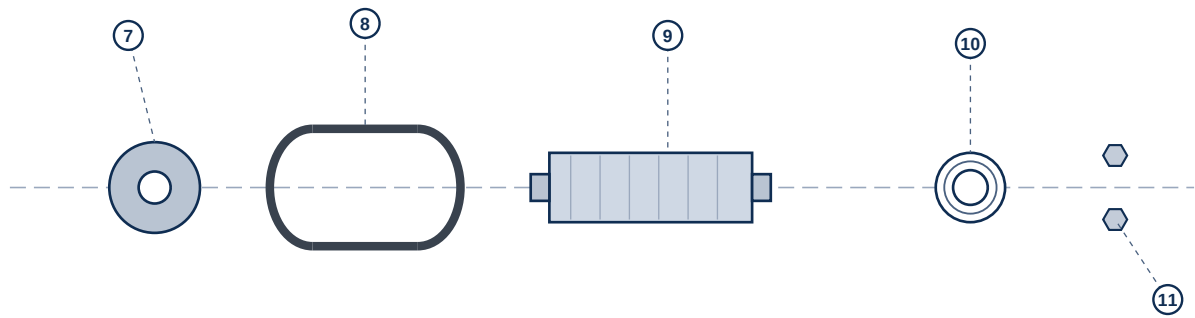
If the checks above do not resolve the symptom, the following parts are the likely cause, most likely first: incline motor, motor controller.

SECTION 6 — ERROR CODES

Codes are shown on the console. Record the code before power cycling — it is not retained across a restart.

Code	Meaning	First actions
E1	Motor controller communication fault. The console has lost communication with the motor control board.	Power the unit off at the wall for 60 seconds, then restart. If the code returns, the console-to-controller data cable or the controller itself has failed.
E2	Speed sensor signal lost. No pulse detected from the speed sensor while the motor is driven.	Check that the sensor is seated against the front roller pulley and that its connector is fully home on the controller.
E5	Incline calibration failure. The incline motor did not reach its home position within the expected travel time.	Run the incline calibration routine from the service menu. If it fails again the incline motor or its limit switch has failed.
E7	Motor over-current. The controller shut the motor down because current exceeded the safe limit — most often excessive belt friction, not a failed motor.	STOP USE. Check belt tension and lubrication first; an over-tight or dry belt is the usual cause. Do not restart repeatedly.
E9	Safety key not detected. The safety key magnet is not registering.	Reseat the safety key. If the unit still will not start, the key magnet has weakened and the key needs replacing.

SECTION 7 — PARTS LIST



Shown exploded along the drive axis. Reassemble in reverse order.

Figure 4 — Drive assembly, exploded view

#	Part number	Description	Cust. fit
7	FF-TT-SUMMIT-350-S-MOTOR	Drive motor and pulley assembly	No
8	FF-TT-SUMMIT-350-S-BELT	Running belt, 500 mm x 1 550 mm	Yes
9	FF-TT-SUMMIT-350-S-FRONT-ROLLER	Front roller, crowned, 60 mm	Yes
10	FF-TT-SUMMIT-350-S-REAR-ROLLER	Rear roller with sealed bearings	Yes
11	FF-TT-SUMMIT-350-S-DECK	Reversible running deck	Yes

Full parts list

Quote the complete part number when ordering. Parts marked technician only must not be fitted by the customer; doing so ends the remaining coverage on the machine.

Part number	Description	Class	Cust. fit
FF-TT-SUMMIT-350-S-BELT	Running Belt	Consumable	Yes
FF-TT-SUMMIT-350-S-DECK	Running Deck	Mechanical	Yes
FF-TT-SUMMIT-350-S-MOTOR	Drive Motor	Electronics	No
FF-TT-SUMMIT-350-S-MOTOR-CONTROLLER	Motor Control Board	Electronics	No
FF-TT-SUMMIT-350-S-CONSOLE	Console Assembly	Electronics	Yes
FF-TT-SUMMIT-350-S-SPEED-SENSOR	Speed Sensor	Electronics	Yes
FF-TT-SUMMIT-350-S-INCLINE-MOTOR	Incline Motor	Mechanical	Yes
FF-TT-SUMMIT-350-S-FRONT-ROLLER	Front Roller	Mechanical	Yes
FF-TT-SUMMIT-350-S-REAR-ROLLER	Rear Roller	Mechanical	Yes
FF-TT-SUMMIT-350-S-SAFETY-KEY	Safety Key	Consumable	Yes
FF-TT-SUMMIT-350-S-POWER-CORD	Power Cord	Consumable	Yes
FF-TT-SUMMIT-350-S-BELT-LUBE	Silicone Belt Lubricant	Consumable	Yes

SECTION 8 — WARRANTY

Component group	Coverage from date of purchase
Frame	120 months
Mechanical parts	36 months
Electronics	24 months
Labour	12 months
Wear and consumable items	90 days, manufacturing defects only

Coverage begins on the date of purchase recorded against the serial number. Commercial or institutional use voids parts and labour coverage under the residential warranty. Wear items — belts, decks, rollers, straps, pads and lubricant — are covered for manufacturing defects for 90 days from purchase and are not covered thereafter.

To make a claim, contact FitForge support with the serial number and the order number. Retain proof of purchase; coverage attaches to the purchase record rather than to the owner.